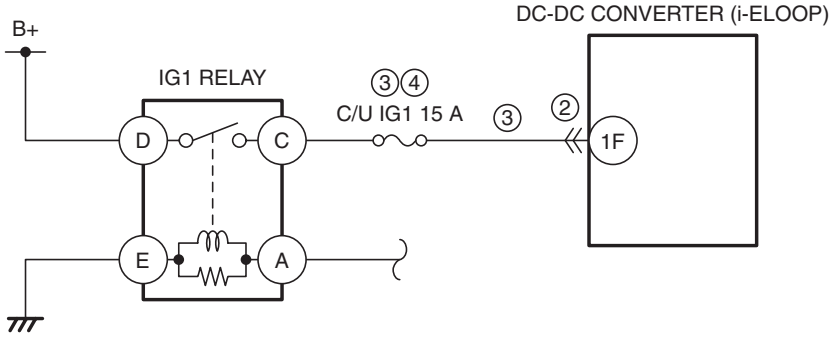
DTC P1628:11 [i-ELOOP]

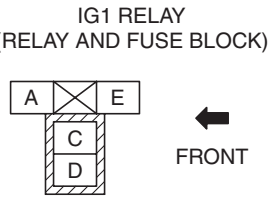
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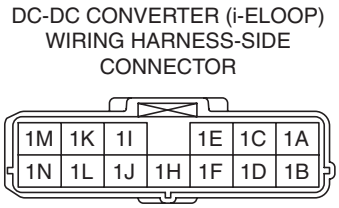
Note

- If the capacitor (i-ELOOP) voltage is low after replacing the DC-DC converter (i-ELOOP), the i-ELOOP indicator light (green) flashes until the capacitor (i-ELOOP) voltage is **approx. 9.5 V**.

DTC P1628:11	**DC-DC converter (i-ELOOP) system: power supply voltage low input**
DETECTION CONDITION	• While the DC-DC converter (i-ELOOP) is receiving CAN signals from the PCM, an IG signal Low condition has continued for **5 s**.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• DC-DC converter (i-ELOOP) connector or terminals malfunction • Short to ground or open circuit in DC-DC converter (i-ELOOP) power supply circuit — Short to ground in wiring harness between IG1 relay terminal C and DC-DC converter (i-ELOOP) terminal 1F — C/U IG1 15 A fuse malfunction — Open circuit in wiring harness between IG1 relay terminal C and DC-DC converter (i-ELOOP) terminal 1F • Short to ground in wiring harness between electric load (such as start stop unit or non-genuine part) connected to C/U IG1 15 A fuse and C/U IG1 15 A fuse, or short to ground in electric load part • DC-DC converter (i-ELOOP) malfunction







Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
2	**INSPECT DC-DC CONVERTER (i-ELOOP) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Disconnect the DC-DC converter (i-ELOOP) connector. • Inspect for poor connection (such as damaged/pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 5.
		No	Go to the next step.
3	**INSPECT DC-DC CONVERTER (i-ELOOP) POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Verify that the DC-DC converter (i-ELOOP) connector is disconnected. • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the DC-DC converter (i-ELOOP) detecting an open circuit. • Measure the voltage at the DC-DC converter (i-ELOOP) terminal 1F (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the C/U IG1 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay terminal C and DC-DC converter (i-ELOOP) terminal 1F. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay terminal C and DC-DC converter (i-ELOOP) terminal 1F. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 5.
4	**VERIFY CONDITION OF C/U IG1 15 A FUSE AND WIRING HARNESS CONDITION RELATED TO ELECTRICAL LOAD PART CONNECTED TO C/U IG1 15 A FUSE** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify the condition of C/U IG1 15 A fuse and wiring harness condition related to electrical load part connected to C/U IG1 15 A fuse. • Is there any malfunction?	Yes	Repair or replace the malfunction part, then go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) • Switch the ignition off. • Switch the ignition ON (engine on) and wait for **5 s or more**. • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) — Start the engine and leave for **100 s or more**. — Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) Go to the next step.
		No	Go to the next step.
6	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	DTC troubleshooting completed.